Machine-checked first-fall-degree reasoning in Weil-descent ECDLP

Anonymous

31 October 2026

Abstract

Subexponential estimates for the elliptic-curve discrete logarithm problem over small-characteristic fields rest on the claim that the polynomial systems from Weil descent have a small *first fall degree*. Several arguments in this line replace the true first fall degree with a quantity that is easier to compute, and the replacement is where errors enter. We give three results, each checked in Lean 4 with Mathlib, and we state for each exactly what the kernel checked and what it did not. (i) Lemma 4 of Nagao (ePrint 2015/984) asserts that the first fall degree is at most a "fake" first fall degree computed modulo the field equations. Under the equality reading of the paper's definition, we give two systems over $\mathbb{F}_2$, in two and three variables, with true first fall degree 3 and fake first fall degree 2. A short argument shows when the inference can be recovered for a given instance, and the paper's own applications satisfy it. (ii) We formalize Theorem 2.8 of Caminata and Gorla: if a system has a degree fall and its last fall degree is finite, the last fall degree is the largest degree at which a fall occurs. A single exhibited fall therefore gives a lower bound and no upper bound. (iii) On a fixed Boolean Semaev S_3 system over $\mathbb{F}_{2^4}$, the direct Macaulay-row statistic and the degree-bounded closure agree on the first degree with a new class but disagree on its multiplicity: 1 new class against 12 at degree three. Over 80 archived draws they agree on the grade of every draw and disagree on the multiplicity for 77. The same audit withdraws an earlier internal inference that every such system falls at degree two. Nothing here gives an ECDLP algorithm or changes a published complexity estimate.

1 Introduction

Semaev's summation polynomials [8] turn a point decomposition on an elliptic curve over $\mathbb{F}_{q^n}$ into a polynomial system. Weil descent turns that system into one over $\mathbb{F}_q$ in more variables. Gröbner-basis algorithms solve it in time governed by the degree they must reach, and the *first fall degree*, introduced in the HFE setting by Dubois and Gama [2], became a common proxy for that degree. Petit and Quisquater [7] conjectured that first fall degree tracks solving degree for these systems. That conjecture, and its variants, underlie the subexponential estimates of Semaev [9] and Nagao [5, 6]. Huang, Kosters and Yeo [3] and Kosters and Yeo [4] gave evidence against the assumption, and Caminata and Gorla [1] put the related invariants on a rigorous footing.

The arguments in this area pass between several quantities that differ slightly: falls measured before or after reduction by the field equations; field equations as generators of the system or as relations of the ring; row spans of a Macaulay matrix or their multiplicative closure. Each step is easy to get wrong and hard to check by eye. We used a proof assistant to check three such steps. One check refutes a published lemma under a stated reading. The second formalizes a known theorem in a form that makes the direction of valid inference explicit. The third shows that two statistics with the same name agree on one number and disagree on another.

What the kernel checked. Every Lean statement cited here builds against a pinned Mathlib and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. No `sorry`, `admit`, custom axiom or `unsafe` declaration appears. Section 6 gives each statement's hosted CI receipt and the status of its independent statement review, which is a separate check that the Lean proposition says what the prose claims. The last-fall theorem, the concrete Semaev witness and the trace identity of Section 5 have passed that review at their stated scope. The counterexample certificate and the Semaev rank family have not.

What this paper does not claim. We say nothing about Nagao's Proposition 5 or about Semaev's first-fall assumption in general. We give no complexity estimate and no algorithm, and no measurement here bears on ECDLP cost at any cryptographic size.

2 Definitions

Let k be a field and $R = k[X_1, \dots, X_N]$. Write deg for total degree.

Definition 1 (First fall degree, equality reading)**.** Let $S = \{s_1, \dots, s_M\} \subset R$. A *fall at degree D* is a tuple $(g_1, \dots, g_M)$ of polynomials with

1. $\deg(g_i s_i) \le D$ for every i, with equality for at least one i;
2. $\sum_i g_i s_i \neq 0$;
3. $\deg(\sum_i g_i s_i) < D$.

The *first fall degree* $D_{\mathrm{ff}}(S)$ is the least $D \ge 1$ at which a fall exists.

Over $\mathbb{F}_p$ let $S_{\mathrm{fe}} = \{X_i^p - X_i\}$ and write $\mathrm{red}(h)$ for the reduction of h modulo S_{fe}, which over $\mathbb{F}_2$ clamps every exponent to at most one.

Definition 2 (Fake first fall degree)**.** $D'_{\mathrm{ff}}(S)$ is defined as $D_{\mathrm{ff}}(S)$ except that cofactors are reduced (multilinear over $\mathbb{F}_2$), every product and the sum are replaced by their reductions, generators that reduce to zero are dropped, and condition (2) asks for a nonzero reduced sum.

Remark 3 (Which reading)*.* Definitions 5 and 6 of [6] print condition (1) as $\max_i \deg(g_i s_i) \ge D$. Under that literal reading, padding the cofactors makes condition (1) free. The first fall degree then collapses to one more than the least degree of a nonzero element of the ideal, and Lemma 4 holds for reasons unrelated to the paper's argument. Nagao's earlier paper [5] prints equality in its Definition 1, and the standard definition used in [1] is the degree at which the fall happens. We adopt the equality reading throughout. It is a declared interpretation of an ambiguous text. We make no claim about the author's intent.

Definition 4 (Caminata–Gorla [1], Definition 1.5)**.** For $F \subset R$ and $i \in \mathbb{N}$, let $V_{F,i}$ be the smallest k-subspace of R that contains every $f \in F$ with $\deg f \le i$ and is closed under $f \mapsto fg$ whenever $f \in V_{F,i}$ and $\deg f + \deg g \le i$. For f in the ideal (F) let $d_f = \min\{d : f \in V_{F,d}\}$, so $d_f \ge \deg f$. Say f *has a degree fall* if $d_f > \deg f$. The *last fall degree* is

$$d_F = \min\{d : \ f \in V_{F,\max(d,\deg f)} \text{ for every } f \in (F)\}.$$

The Lean formalization writes the closure condition as $\deg f + \deg g \le i$ rather than $\deg g \le i - \deg f$. On natural numbers the second form truncates and would admit degree-zero multipliers of any f with $\deg f > i$.

3 A counterexample to Lemma 4 of ePrint 2015/984

Lemma 4 of [6] states $D_{\mathrm{ff}} \le D'_{\mathrm{ff}}$, so that a bound on the easily computed fake quantity would bound the true one. Two placements of S_{fe} are possible on the true side. The printed lemma takes D_{ff} of $\{f_1, \ldots, f_M\}$ alone. The *inside form* takes both quantities of the same system $S = F \cup S_{\mathrm{fe}}$, which is the form the paper's arguments need. Lean checks the refutation of the inside form. The printed form fails on the same witnesses: without S_{fe}, $X^2Y + X(XY + X) = X^2$ and $X_3(X_1X_2 + X_3) + X_2(X_1X_3 + X_2) = X_2^2 + X_3^2$ are falls at degree 3, and the degree-2 argument below applies unchanged. That check is by hand and by exhaustive search, not in Lean.

Theorem 5. *Over $\mathbb{F}_2$, with S_{fe} among the generators of the true system, let*

$$\begin{aligned} &(A) \quad F_A = \{X^2Y,\ XY + X\} \subset \mathbb{F}_2[X, Y], \\ &(B) \quad F_B = \{X_1X_2 + X_3,\ X_1X_3 + X_2\} \subset \mathbb{F}_2[X_1, X_2, X_3]. \end{aligned}$$

For $S = F_A \cup S_{\mathrm{fe}}$ and for $S = F_B \cup S_{\mathrm{fe}}$ we have $D_{\mathrm{ff}}(S) = 3$ and $D'_{\mathrm{ff}}(S) = 2$. Hence $D_{\mathrm{ff}} \le D'_{\mathrm{ff}}$ fails, and the universal statement over finite systems over $\mathbb{F}_2$ is false.

Proof. Fake side. For (A), $\mathrm{red}(X^2Y) = XY$, and $\mathrm{red}(1 \cdot X^2Y) + \mathrm{red}(1 \cdot (XY + X)) = X$ has degree $1 < 2$, so there is a fake fall at degree 2. For (B), $\mathrm{red}((X_1 + 1)(X_1X_3 + X_2)) = X_1X_2 + X_2$, and adding $X_1X_2 + X_3$ gives $X_2 + X_3$, again a fake fall at degree 2. No fake fall exists at degree 1, because every reduced generator has degree 2.

True side, degree 3. For (A), $X^2Y + X \cdot (XY + X) = X^2$. For (B), $X_3\,(X_1X_3 + X_2) + X_1\,(X_3^2 + X_3) = X_2X_3 + X_1X_3$. Each is a fall at degree 3.

True side, no fall below 3. At $D = 1$ every generator has degree at least 2, so no product has degree exactly 1. At $D = 2$ a product of degree at most 2 needs a constant cofactor on a quadratic generator and the zero cofactor on a cubic one. A fall then needs the degree-2 parts to cancel. For (A) those parts are XY, X^2, Y^2, and for (B) they are X_1X_2, X_1X_3, X_1^2, X_2^2, X_3^2. Both sets are linearly independent, so no fall exists. □

The Lean module `CryptoResearch.Nagao.Lemma4Witness` checks all of these facts by `decide +kernel` in a model where a polynomial over $\mathbb{F}_2$ is a duplicate-free list of exponent vectors and cofactors range over polynomials whose monomials have degree at most D (multilinear on the fake side). The named theorem `lemma4_inside_false` states $\neg\, \forall n\, S,\ \mathrm{Lemma4Inside}\ n\ S$. A separate Python implementation, written from the definitions without reference to the first, reproduces both witnesses. It also finds the lemma violated on 94 of 400 random pairs of multilinear quadratics in three variables (23.5%) [10]. We reproduced both witnesses a third time by exhaustive search over cofactor tuples while preparing this paper.

3.1 When the inference survives

The failure has one cause. Reduction modulo S_{fe} can lower the degree of an individual product g_is_i, so a fake fall need not come from any true fall at the same degree. The paper's Lemma 3 controls the degree of the sum and says nothing about the products whose maximum defines D_{ff}. The following proposition makes the repair explicit.

Proposition 6 (Instance-wise recovery)**.** *Over $\mathbb{F}_2$, let $S = F \cup S_{\mathrm{fe}}$ and let (g_i) be a fake fall at degree v for the generators in F. If every unreduced product satisfies $\deg(g_if_i) \le v$, then $D_{\mathrm{ff}}(S) \le v$.*

Proof. Let $s = \sum_i g_i f_i$ and $r = \mathrm{red}(s)$, so $r \neq 0$ and $\deg r < v$. Reduce s one monomial at a time, replacing $X_j^2 m$ by $X_j m$. Each step adds $m(X_j^2 + X_j)$, whose degree is at most the degree of the monomial being reduced, and degrees never increase along the way. So $r = s + \sum_j h_j(X_j^2 + X_j)$ with $\deg(h_j(X_j^2 + X_j)) \leq \deg s \leq v$. This is a combination of the generators of S in which every product has degree at most v and the sum r is nonzero of degree less than v. Some reduced product $\mathrm{red}(g_i f_i)$ has degree exactly v, and the unreduced product has at least that degree, so by hypothesis it has degree exactly v. Hence there is a fall at degree v, and $D_{\mathrm{ff}}(S) \leq v$. □

Proposition 6 is not formalized in Lean; the proof above is the only check. It needs two hypotheses that neither [5] nor [6] states as such: the field equations are generators of the true system, and the fake witness loses no degree under reduction. In both witnesses of Theorem 5 the second hypothesis fails: X^2Y drops from degree 3 to 2, and $(X_1 + 1)(X_1X_3 + X_2)$ drops from 3 to 2.

According to our reading of the source, the only use of Lemma 4 in [6] is in the proof of its Lemma 6. That proof puts S_{fe} on the true side and tracks product degrees with explicit cofactors of degree at most one on reduced generators. Both hypotheses of Proposition 6 can be checked there. [5] does not state Lemma 4 and counts the field equations among its $n + N$ equations. In our reading, the published uses survive. What fails is the lemma as a general tool: a later argument that cites Lemma 4 for a fake-degree bound is relying on a false statement.

4 The last fall degree is the largest fall

Theorem 7 (Caminata–Gorla [1], Theorem 2.8; formalized)**.** *Let k be a field and $F \subset k[X_1, \ldots, X_N]$. Suppose d_F is finite and some $f \in (F)$ has a degree fall. Then*

$$d_F = \max\{\, d_f : \ f \in (F),\ d_f > \deg f \,\}.$$

The Lean statement

`CryptoResearch.FallDegree.lastFallDegree_eq_max_degreeFall`

is over Mathlib's `MvPolynomial` for an arbitrary variable type and arbitrary field. It asserts that d_F is the greatest element of the set of fall degrees. Finiteness of d_F is an explicit hypothesis: Caminata and Gorla obtain it from the existence of a Gröbner basis, and we did not reprove that. The hypothesis that a fall exists is also explicit. The formalization does not construct a system satisfying both. The source review works through $F = \{x^2,\ x^2 + x\}$, whose last fall degree is 2, on paper only. The same review noted a local wording slip ("$f \in F$") in the published proof and supplied a direct derivation; the theorem is unaffected.

The module also proves the lower-bound direction on its own: if some f in the ideal has $d_f > \deg f$, then $d_F \geq d_f$. This is the direction an experiment can use. Observing one fall certifies that the last fall degree is at least the degree of that fall. It certifies nothing in the other direction. An upper bound needs a closure argument covering every element of the ideal, which a finite computation of a Macaulay rank profile does not provide unless it is proved to compute the spaces $V_{F,i}$ and some bound rules out later falls. Lemma 4 of [6] was an attempt to get an upper bound cheaply, in the first-fall setting. Section 3 shows that the shortcut fails.

5 Two first-fall statistics on a Boolean Semaev system

The third result is about finite computation. It shows how two statistics, both reasonably called "first fall", separate on a real system.

The system. Take the curve $y^2 + xy = x^3 + x^2 + 1$ over $\mathbb{F}_{2^4} = \mathbb{F}_2[t]/(t^4 + t + 1)$ and the third summation polynomial $S_3 = e_2^2 + e_3 + 1$ in the elementary symmetric functions of three abscissae. Fix the target abscissa $x_R = 1$ and restrict the two free abscissae to an $\mathbb{F}_2$-subspace with encoded basis $[9, 3, 2, 12]$. Weil descent yields four Boolean equations in eight variables $u_0, \ldots, u_7$, in $\mathbb{F}_2[u]/(u_i^2 - u_i)$. The archived equations have SHA-256 `89ab8e66...ba6eb` [10].

Two statistics. The *direct-row* statistic at degree D is the number of new low-degree classes, $\dim(\mathrm{rows}_D \cap R_{<D}) - \dim \mathrm{rows}_{D-1}$, where rows_D is the span of the multiplier-times-equation rows of degree at most D. The *closure* statistic replaces rows_D by the least subspace containing the equations and closed under multiplication within the degree budget, the Boolean-quotient analogue of $V_{F,D}$.

Theorem 8 (finite certificates)**.** *For this system the direct-row ranks at degrees* $1, 2, 3$ *are* $1, 11, 53$*, with top-degree projection ranks* $1, 10, 41$*, so the direct-row new-class counts are* $0, 0, 1$*. There is an explicit vector in* rows_3 *of degree at most two that lies outside* rows_2*.*

Lean checks these ranks and the witness. The witness is an XOR of selected degree-three rows, whose coefficient sum at monomial masks 80 and 192 vanishes on every degree-two row but equals one on the witness. The modules are `CryptoResearch.Semaev.BooleanParity*` and `CryptoResearch.Semaev.ConcreteFirstFallWitness`, and they also check that the four transcribed equations match a direct evaluation of S_3 on all 256 Boolean assignments.

An independent code review then computed the closure statistic on the same system and, separately, on all 80 archived draws [10]. Three fixed-point methods agree on the closed spaces, checked by joint rank.

D	direct rank / top / low	closed rank / top / low	new classes (direct, closed)
1	1 / 1 / 0	1 / 1 / 0	–
2	11 / 10 / 1	11 / 10 / 1	0, 0
3	53 / 41 / 12	79 / 56 / 23	1, 12

On this draw the two statistics agree that the first new class appears at degree three and disagree on how many appear: 1 against 12. Across the 80 draws they agree on the first degree with a positive count for every draw (68 at degree two, 12 at degree three) and disagree on the count at degree three for 77 draws. At degrees one and two, seven direct-row spaces contain rows outside the closure. The direct-row policy admits a product based on its degree after cancellation, even when the multiplier exceeds the degree budget of the closure definition.

A withdrawn inference. An earlier internal note proved in Lean a trace identity: for $x_R \neq 0$ and $\mu = x_R^{-2}$, one has $\mu^{1/2} + \mu x_R = 0$ in $\mathbb{F}_{2^n}$ (`trace_witness_annihilates_degree_two_part`). It cited the identity for a nontrivial first fall at degree two for every admissible parameter choice. The identity is correct. The inference does not follow, because the theorem contains no statement about membership in a lower row space, and the zero map satisfies its premises. The draw above has $x_R \neq 0$ and no new class at degree two, which refutes the uniform claim in this finite model. The same review found that the historical counting code combined Boolean monomials by set union where coefficients over $\mathbb{F}_2$ must cancel in pairs; correcting this changes the archived split of first-fall grades over the 80 draws from 70/10 to 68/12. Kosters and Yeo [4] describe degree-two first falls as usual, which is consistent with 68 of 80.

6 Verification status

Statement	Kernel	Independent statement review
Thm. 5 (`lemma4_inside_false`)	hosted receipt, CI run 41155781140	not yet reviewed; producer caps fidelity at partial (two scoping rows)
Prop. 6	not formalized	paper proof only
Thm. 7	hosted receipt, CI run 41191503149	validated at stated scope (three-joint review)
Thm. 8, concrete witness	hosted receipt, CI run 41155781140	validated at scope, one draw
Thm. 8, rank family	hosted receipts, CI run 41155781140	integrity audit only
Trace identity (§5)	hosted receipt, CI run 41155781140	identity valid; uniform inference withdrawn
Closure counts (§5)	not formalized	independent code review

A hosted receipt records that the workspace built, the per-theorem audit found the named declaration with only the three standard axioms, and the target specification and module hashes matched the frozen inventory. Each receipt's own field for semantic review reads *pending*; reviews are filed separately and are summarized in the last column.

7 Discussion

The three results point one way. In first-fall arguments the dangerous step is moving between a quantity that is easy to measure and the one that matters: reduced against unreduced products, row spans against their closure, one observed fall against a bound on all of them. A kernel catches a false lemma only once someone writes the lemma down precisely enough to check. An independent statement review catches a true theorem that is cited for more than it says. We needed both: the first for Theorem 5, the second for the withdrawn trace inference.

Three items would strengthen this note. The first is a Lean proof of the printed (outside) form of the counterexample and of Proposition 6. The second is a bridge from the list-of-monomials model to `MvPolynomial`. The third is an independent review of the Lean statement of Theorem 5.

References

[1] A. Caminata and E. Gorla. Solving degree, last fall degree, and related invariants. *J. Symbolic Comput.*; IACR ePrint 2021/1611; arXiv:2112.05579.

[2] V. Dubois and N. Gama. The degree of regularity of HFE systems. In *ASIACRYPT 2010*, LNCS 6477, 2010.

[3] M.-D. Huang, M. Kosters, and S. L. Yeo. Last fall degree, HFE, and Weil descent attacks on ECDLP. In *CRYPTO 2015*; IACR ePrint 2015/573.

[4] M. Kosters and S. L. Yeo. Notes on summation polynomials. arXiv:1503.08001, 2015.

[5] K. Nagao. Equations system coming from Weil descent and subexponential attack for algebraic curve cryptosystem. IACR ePrint 2013/549.

[6] K. Nagao. Complexity of ECDLP under the first fall degree assumption. IACR ePrint 2015/984.

[7] C. Petit and J.-J. Quisquater. On polynomial systems arising from a Weil descent. In *ASIACRYPT 2012*, LNCS 7658, 2012.

[8] I. Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. IACR ePrint 2004/031.

[9] I. Semaev. New algorithm for the discrete logarithm problem on elliptic curves. IACR ePrint 2015/310; arXiv:1504.01175.

[10] Artifact repository for this note (anonymized for review). Lean sources under `formal/CryptoResearch/`; counterexample certificate `tools/lemma4_inside_form.py`; closure audit `docs/reports/2026-10-26-semaev-closure-review/`; hosted receipts `docs/reports/2026-10-31-hosted-formal-receipts/`.